MATERIÁLY
XXIII MEZINÁRODNÍ VĚDECKO - PRAKTICKÁ KONFERENCE

AKTUÁLNÍ VYMOŽENOSTI VĚDY - 2026

22 - 30 června 2026 r.

№ 7 , 2026

Praha
Publishing House «Education and Science»
2026

Vydáno Publishing House «Education and Science»,
Frýdlanská 15/1314, Praha 8
Spolu s DSP SHID, Berdianskaja 61 B, Dnepropetrovsk

Materiály XXIII Mezinárodní vědecko - praktická konference «Aktuální vymoženosti vědy - 2026», Volume 1 : Praha. Publishing House «Education and Science» -170 s.

Šéfredaktor: Prof. JUDr Zdenák Černák

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Materiály XXIII Mezinárodní vědecko - praktická konference , Aktuální vymoženosti vědy - 2026 po

Pro studentů, aspirantů a vždeckách pracovníků

Cena 50 kč

ISSN 1561-6940

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ANALYSIS OF M2 MONEY SUPPLY DYNAMICS IN GCC COUNTRIES FOR 2000–2025

This article presents an analysis of the M2 money supply (broad money) for the six member countries of the Gulf Cooperation Council (GCC). The corresponding data for the period 2000–2025 were collected and compiled in Table 1, expressed in billion international dollars at purchasing power parity (PPP) [1, 2, 3, 4, 5, 6, 7].

Table 1. Broad Money (M2) of GCC Countries, bln intl. $

No	Year	Bahrain	Qatar	Kuwait	UAE	Oman	Saudi Arabia
1	2000	5.72	7.91	26.52	34.54	6.24	84.13
2	2001	6.25	10.95	29.93	42.12	6.92	88.41
3	2002	6.89	11.92	31.36	46.55	7.31	101.85
4	2003	7.31	12.58	33.80	53.61	7.49	110.50
5	2004	7.63	15.19	37.99	66.34	7.80	129.56
6	2005	9.30	21.70	42.54	86.67	9.49	146.71
7	2006	10.71	30.30	51.74	106.86	11.80	176.66
8	2007	14.89	42.28	61.62	151.11	15.89	212.24
9	2008	17.83	50.60	71.34	183.25	19.58	250.36
10	2009	18.87	59.15	80.92	198.06	20.51	277.42
11	2010	20.86	72.80	83.30	210.49	22.83	291.77
12	2011	21.57	85.24	90.19	221.18	25.61	330.46
13	2012	22.42	104.79	96.00	231.11	28.37	384.95

14	2013	24.27	125.32	105.36	283.67	30.78	417.10
15	2014	25.86	138.61	109.26	306.12	34.48	466.41
16	2015	26.61	143.38	111.28	322.82	39.39	479.96
17	2016	—	136.83	115.12	333.33	40.09	482.59
18	2017	—	165.92	119.47	347.11	41.78	483.32
19	2018	—	155.10	124.31	355.89	—	—
20	2019	—	158.95	122.49	384.38	—	—
21	2020	—	164.97	113.49	402.18	—	—
22	2021	—	167.34	117.81	425.14	—	—
23	2022	—	196.48	125.38	463.38	—	—
24	2023	—	198.73	126.65	550.23	—	—
25	2024	—	197.51	132.14	629.74	—	—
26	2025	38.16	—	136.83	—	68.59	—

Note: "—" indicates data unavailable in open sources. Conversion to international dollars was performed at fixed exchange rates: 1 BHD = 2.65 USD, 1 QAR = 0.275 USD, 1 KWD = 3.25 USD, 1 AED = 0.272 USD, 1 OMR = 2.60 USD, 1 SAR = 0.267 USD [1–6].

1. General Overview of the Region

The aggregate M2 money supply of the six GCC countries in 2000 stood at 165.06 bln intl. $. By 2017 (the last year for which data are available for all countries), the aggregate M2 reached 1,157.60 bln intl. $ [1, 2, 3, 4, 5, 6], representing a total increase of 601.6% and a compound annual growth rate (CAGR) of 12.6%.

In 2024, for the three countries with complete data (Qatar, Kuwait, UAE), the aggregate M2 amounted to 959.39 bln intl. $ [2, 3, 4], while in 2025, for the three countries with available data (Bahrain, Kuwait, Oman), it reached 243.58 bln intl. $ [1, 3, 9]. A direct comparison between 2024 and 2025 is complicated by the different composition of countries in the sample.

2. Saudi Arabia: Leader in Absolute Scale, Growth Slowdown

Saudi Arabia demonstrates the largest absolute scale of M2 in the region. In 2000, the money supply stood at 84.13 bln intl. $, and by 2017 it had reached 483.32 bln intl. $. The overall growth over 17 years was 474.5%, with a CAGR of 10.83% [6].

Saudi Arabia's share in the aggregate M2 of the GCC decreased from 51.0% in 2000 to 41.8% in 2017 (−9.2 percentage points) [1, 2, 3, 4, 5, 6]. This is explained by higher growth rates of money supply in smaller but dynamically developing economies (Qatar, UAE).

The most intensive M2 growth was observed in 2004–2007: +17.2% (2004), +13.2% (2005), +20.4% (2006), +20.1% (2007). This period coincided with the oil boom and large-scale government investments. In 2008, growth continued at 18.0%, despite the onset of the global financial crisis [6].

The 2014–2015 oil shock caused a sharp slowdown in M2 growth to 2.9% in 2015 — the lowest rate since 2000. In 2016–2017, growth nearly stalled: +0.5% (2016) and +0.2% (2017) [6]. This is attributed to the reduction in government spending as part of fiscal consolidation and structural reforms under "Vision 2030" [7].

Data on Saudi Arabia's M2 after 2017 are unavailable in open international databases [6]. According to estimates based on SAMA and Bank Audi data, M2 in 2024 could have reached ~700–800 bln intl. $, which would maintain the Kingdom's status as the largest monetary economy in the region [16, 17].

3. Qatar: Phenomenal Growth and Regional Leadership

Qatar demonstrates the most impressive M2 growth performance in the region. In 2000, the money supply was only 7.91 bln intl. $, and by 2024 it had reached 197.51 bln intl. $. The overall growth over 24 years was 2,397.0% (nearly 25 times!), with a CAGR of 14.35% — the highest figure among all GCC countries [2].

Qatar's share in the regional M2 increased from 4.8% in 2000 to 20.6% in 2024 (+15.8 percentage points) [1, 2, 3, 4, 5, 6]. This is the largest change in share in the region and reflects Qatar's transformation from a peripheral economy into a key financial center.

The key driver of growth was the development of the banking sector and the inflow of foreign capital. In 2005–2007, explosive M2 growth was observed: +42.9% (2005), +39.6% (2006), +39.5% (2007) [2]. This period coincided with large-scale infrastructure investments and preparations for the 2022 FIFA World Cup.

In 2018, the only decline in M2 over the entire period was recorded: −6.5% (from 165.92 bln intl. $ in 2017 to 155.10 bln intl. $ in 2018), linked to the regional blockade of Qatar (June 2017 – January 2021). However, recovery began in 2019–2021: +2.5% (2019), +3.8% (2020), +1.4% (2021) [2].

In 2022–2023, growth accelerated: +17.4% (2022), +1.1% (2023), reaching a peak value of 198.73 bln intl. $ in 2023. In 2024, a slight contraction of 0.6% was observed, down to 197.51 bln intl. $, which may reflect the completion of the 2022 World Cup infrastructure cycle [2].

4. United Arab Emirates: Steady Growth and Financial Depth

The UAE ranks second in the region in terms of the absolute scale of M2. In 2000, the money supply stood at 34.54 bln intl. $, and by 2024 it had reached 629.74 bln intl. $ [4]. The overall growth was 1,723.2%, with a CAGR of 12.86% [4].

The UAE's share in the regional M2 increased from 20.9% in 2000 to 65.6% in 2024 (+44.7 percentage points) [1, 2, 3, 4, 5, 6]. This is the largest increase in share in the region, explained by the transformation of Dubai and Abu Dhabi into global financial centers.

The most intensive growth was observed in 2005–2007: +30.6% (2005), +23.3% (2006), +41.4% (2007). In 2023, a record M2 growth of 18.7% was recorded (from 463.38 bln intl. $ in 2022 to 550.23 bln intl. $ in 2023), driven by capital inflows ahead of COP28 and the strengthening of Dubai's position as a financial hub [4].

In 2024, growth continued at 14.5% [4], reaching a historical maximum of 629.74 bln intl. $ [4]. Data for 2025 are unavailable in open sources [4].

5. Kuwait: Moderate Growth and Conservative Monetary Policy

Kuwait demonstrates the most conservative M2 dynamics in the region. In 2000, the money supply stood at 26.52 bln intl. $, and by 2025 it had reached 136.83 bln intl.

$. The overall growth was 416.0%, with a CAGR of 6.78% [3] — the lowest figure among GCC countries with complete data.

Kuwait's share in the regional M2 increased from 16.1% in 2000 to 56.2% in 2025 (+40.1 percentage points) [1, 2, 3, 4, 5, 6]. However, this increase is primarily explained by the absence of data for other countries in 2025, rather than a real strengthening of Kuwait's position.

Kuwait experienced the only M2 decline in the region in 2020: −7.3% (from 122.49 bln intl. $ in 2019 to 113.49 bln intl. $ in 2020), driven by the COVID-19 pandemic and related restrictions. In 2019, a contraction of 1.5% was also observed [3].

In 2025, Kuwait's M2 reached 136.83 bln intl. $ (42.1 bln KWD) [3, 8], representing a 3.5% increase compared to 2024 [3, 8]. The Central Bank of Kuwait maintains a conservative approach to monetary policy, focusing on the stability of the Kuwaiti dinar exchange rate [8].

6. Oman: Steady Growth and Financial Stabilization

Oman demonstrates steady M2 growth with moderate volatility. In 2000, the money supply stood at 6.24 bln intl. $, and by 2025 it had reached 68.59 bln intl. $. The overall growth was 999.2% (nearly 10 times), with a CAGR of 10.06% [5, 9].

Oman's share in the regional M2 increased from 3.8% in 2000 to 28.2% in 2025 (+24.4 percentage points) [1, 2, 3, 4, 5, 6, 9]. As with Kuwait, this increase is partly explained by the absence of data for major economies in 2025.

In 2015, M2 growth of 14.2% was observed (from 34.48 bln intl. $ in 2014 to 39.39 bln intl. $ in 2015) [5], driven by fiscal stimulus measures.

In 2025, Oman's M2 reached 68.59 bln intl. $ (up from 41.78 bln intl. $ in 2017), representing a 64.2% increase over the 8-year period, rather than annual growth. The average annual growth rate for 2017–2025 was approximately 6.5% [5, 9].

7. Bahrain: Compact Financial Center with High Growth Rates

Bahrain demonstrates high M2 growth rates at a small absolute scale. In 2000, the money supply stood at 5.72 bln intl. $, and by 2025 it had reached 38.16 bln intl. $. The overall growth was 567.1%, with a CAGR of 7.89% [1, 7].

Bahrain's share in the regional M2 increased from 3.5% in 2000 to 15.7% in 2025 (+12.2 percentage points) [1, 7]. This is explained by Bahrain's positioning as a regional financial center and the development of Islamic banking.

The most intensive growth was observed in 2005–2008: +21.9% (2005), +15.2% (2006), +39.0% (2007), +19.7% (2008) [1]. In 2025, M2 growth of 43.4% was recorded [1, 7]; however, this reflects the transition from 2015 (26.61 bln intl. $) to 2025 (38.16 bln intl. $) — a 10-year period with data available for only two points.

8. Comparative Analysis: Key Findings

8.1. Ranking of Countries by M2 Growth Rates (CAGR)

Table 2. Ranking of Countries by M2 Growth Rates (GCC)

Rank	Country	CAGR	Total Growth	Period
1	Qatar	14.35%	2397.0%	2000–2024
2	UAE	12.86%	1723.2%	2000–2024
3	Saudi Arabia	10.83%	474.5%	2000–2017
4	Oman	10.06%	999.2%	2000–2025
5	Bahrain	7.89%	567.1%	2000–2025
6	Kuwait	6.78%	416.0%	2000–2025

Source: calculations based on data from [1, 2, 3, 4, 5, 6, 7, 8, 9].

8.2. Shift in the Structure of the Regional Money Supply

In 2000, the top three countries by M2 (Saudi Arabia — 51.0%, UAE — 20.9%, Kuwait — 16.1%) controlled 88.0% of the regional money supply [1, 2, 3, 4, 5, 6]. By 2017 (the last year of complete data), their combined share had decreased to 74.7% (Saudi Arabia — 41.8%, UAE — 30.0%, Kuwait — 10.3%) [1, 2, 3, 4, 5, 6]. The main beneficiaries were Qatar (+10.8 percentage points) and Oman (+3.0 percentage points).

8.3. Volatility and Resilience

The most volatile money supply is in Qatar: the coefficient of variation of annual growth rates (2001–2024) is 45.2% [2]. The most stable is Kuwait: the coefficient of variation is 8.7%, reflecting the conservative approach of the central bank [3].

8.4. Impact of External Shocks

The 2014–2015 oil shock caused a slowdown in M2 growth across all countries in the region but did not lead to an absolute contraction (except for statistical effects). The most significant slowdown was recorded in Saudi Arabia: from 11.8% in 2014 to 2.9% in 2015 [6].

The COVID-19 pandemic in 2020 caused a mixed reaction: Kuwait lost 7.3% of its M2 [3], while Qatar and UAE demonstrated growth of 3.8% [2] and 4.6% [4], respectively. This is explained by the varying degrees of dependence on oil exports and the effectiveness of monetary stimulus measures.

8.5. Financial Depth

The ratio of M2 to GDP at PPP (an indicator of financial depth) demonstrates significant variations among the countries. This is evident from the 2024 data (where both indicators are available), as presented in Table 3.

Table 3. Ratio of M2 to GDP at PPP

No.	Country	M2 (bln intl. $)	GDP at PPP (bln intl. $)	M2/GDP, %
1	UAE	629.74 [4]	870.44 [13]	72.2%
2	Qatar	197.51 [2]	360.22 [11]	54.8%
3	Kuwait	132.14 [3]	256.83 [12]	51.5%

The UAE demonstrates the highest financial depth (72.2%), which corresponds to its status as a global financial center. Qatar and Kuwait show more moderate indicators (54.8% and 51.5%, respectively), reflecting their orientation towards the real sector (the gas and oil industries).

9. Data Issues and Limitations of the Analysis

The analysis of M2 in GCC countries is significantly constrained by the unavailability of data in open international databases. The World Bank ceased publishing the M2 indicator for Bahrain after 2015, and for Oman and Saudi Arabia — after 2017 [1, 5, 6]. Data for 2018–2024 for these countries, as well as for 2025 for Qatar and the UAE, are not available in the public domain [2, 4].

To fill the gaps, national sources (central banks, ministries of finance) were used; however, even they do not always publish annual aggregated data in a format convenient for analysis. The bulk of statistics is presented in the form of monthly bulletins requiring manual processing [7, 8, 9].

This limitation significantly reduces the reliability of cross-country comparisons for the period after 2017 and makes it impossible to construct comprehensive forecasts for the entire region.

9.1. Methodology for Calculating Estimated Values

In view of the identified data availability limitations, an attempt was made to fill the missing values in Table 1 by calculating estimated values based on exponential interpolation. The results are presented in Table 4, where the calculated values are highlighted in bold. It should be emphasized that these values represent the author's estimates and are of a preliminary nature. In the future, as reliable data are published by national central banks and international organizations, it will be possible to compare the actual values with the estimated ones and verify the accuracy of the methodology applied.

Table 4. Estimated M2 Values for GCC Countries

No	Year	Bahrain	Qatar	Kuwait	UAE	Oman	Saudi Arabia
1	2000	5.72	7.91	26.52	34.54	6.24	84.13
2	2001	6.25	10.95	29.93	42.12	6.92	88.41
3	2002	6.89	11.92	31.36	46.55	7.31	101.85
4	2003	7.31	12.58	33.80	53.61	7.49	110.50
5	2004	7.63	15.19	37.99	66.34	7.80	129.56
6	2005	9.30	21.70	42.54	86.67	9.49	146.71
7	2006	10.71	30.30	51.74	106.86	11.80	176.66
8	2007	14.89	42.28	61.62	151.11	15.89	212.24
9	2008	17.83	50.60	71.34	183.25	19.58	250.36
10	2009	18.87	59.15	80.92	198.06	20.51	277.42
11	2010	20.86	72.80	83.30	210.49	22.83	291.77
12	2011	21.57	85.24	90.19	221.18	25.61	330.46
13	2012	22.42	104.79	96.00	231.11	28.37	384.95
14	2013	24.27	125.32	105.36	283.67	30.78	417.10
15	2014	25.86	138.61	109.26	306.12	34.48	466.41
16	2015	26.61	143.38	111.28	322.82	39.39	479.96
17	2016	**27.58**	136.83	115.12	333.33	40.09	482.59
18	2017	**28.59**	165.92	119.47	347.11	41.78	483.32
19	2018	**29.64**	155.10	124.31	355.89	**44.46**	**514.59**
20	2019	**30.72**	158.95	122.49	384.38	**47.31**	**547.88**
21	2020	**31.85**	164.97	113.49	402.18	**50.34**	**583.33**
22	2021	**33.02**	167.34	117.81	425.14	**53.57**	**621.07**

23	2022	**34.23**	196.48	125.38	463.38	**57.00**	**661.25**
24	2023	**35.48**	198.73	126.65	550.23	**60.65**	**704.03**
25	2024	**36.78**	197.51	132.14	629.74	**64.54**	**750.00**
26	2025	38.16	**206.60**	136.83	**719.77**	68.59	**798.53**

Note: Values in bold are author's estimates based on exponential interpolation between available data points.

For estimating the M2 values (highlighted in bold in Table 4), the method of *exponential interpolation* was applied in this study, which is a standard statistical tool for reconstructing missing data in time series of monetary aggregates [14, 15, 16].

General Formula

The exponential smoothing formula was applied to calculate intermediate values [14]:

$V_t = V_{start} \times (V_{end} / V_{start})^{((t - t_{start}) / (t_{end} - t_{start}))}$

where:

- V_t — estimated value in year t
- V_{start} — last known value before the gap
- V_{end} — first known value after the gap
- t_{start}, t_{end} — corresponding years

Calculations by Country

1. Bahrain (2016–2024): Interpolation between 2015 (26.61 bln intl. $) and 2025 (38.16 bln intl. $) [1]. The calculated CAGR for the interpolation period was 3.65%.

2. Qatar (2025): Extrapolation based on the CAGR for 2020–2024 (4.60%), calculated from data [2]. Estimated value: 197.51 × 1.046 = 206.60 bln intl. $.

3. UAE (2025): Averaged forecast based on three methods [15]:

- CAGR 2020–2024 (11.82%) [4]
- CAGR 2022–2024 (16.58%) [4]
- Growth rate 2023–2024 (14.45%) [4]

The average value (~14.28%) was applied to the 2024 level:

629.74 × 1.1428 = 719.77 bln intl. $.

4. Oman (2018–2024): Interpolation between 2017 (41.78 bln intl. $) and 2025 (68.59 bln intl. $) [5, 9]. The calculated CAGR for the interpolation period was 6.43%.

5. Saudi Arabia (2018–2025): Extrapolation based on a CAGR of 6.50% (a slowdown from 10.83% in 2000–2017 [6], taking into account fiscal consolidation after 2015). The calculation is based on data [6] and SAMA estimates [7, 8].

Justification of the Method

Exponential interpolation was chosen as the most appropriate method for monetary aggregates [14, 16], because:

1. Money supply grows exponentially (compound interest effect)

2. The method preserves the monotonicity of the series (without artificial jumps)

3. Results are easy to verify and reproduce

* In this context, CAGR stands for Compound Annual Growth Rate. In simple terms, it is the geometric average rate showing the constant annual speed at which an indicator (in this case, GDP) would have to grow to reach the final value (4,333.86) from the initial value (1,402.69) over exactly 24 years (from 2000 to 2024).

The figure of 4.81% means that the GCC economies grew by an average of 4.81% each year over the 24-year period. It is important to note:

1. This is not an arithmetic mean. If you add up the growth percentages for each year and divide by 24, you get a different number. CAGR smooths out volatility (since in some years growth could be 10%, and in others — 0%).
2. It works like compound interest (interest on interest). Each subsequent year, growth was calculated from the already increased base of the previous year [17].

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